Wenhan Tang

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EDUCATION

Ph.D. in Atmospheric Sciences, University of Illinois Urbana-Champaign (GPA 4.0) **05/2027 (expected)**
M.S. in Atmospheric Sciences, University of Illinois Urbana-Champaign (GPA 4.0) **05/2024**
B.S. in Atmospheric Sciences, Nanjing University of Information Science & Technology (GPA 3.84, top 6%) **06/2019**

SKILLS

- **Languages & Frameworks:** Python, C/C++, Fortran, SQL, Bash; PyTorch, TensorFlow, scikit-learn
- **Machine Learning:** representation learning, VAEs, permutation-invariant set encoders, Neural ODEs & latent dynamics, RNNs/LSTMs, multistep BPTT, transfer learning, Bayesian inference
- **ML Systems & HPC:** GPU training & batched inference, model/data-pipeline optimization, CUDA, Slurm, MPI/OpenMP, MLOps (Docker, pytest, CI/CD), ETL & API design, NetCDF/xarray, Zarr, Linux/Git, AWS/GCP

PROFESSIONAL EXPERIENCE

Research Assistant, University of Illinois Urbana-Champaign, Urbana, IL, USA **2022 – Present**

➤ ***Neural Surrogate & ML Systems for Particle-Resolved Physics (AeroMELD)*** | *Machine Learning Engineer*

- Architected and released AeroMELD, a neural reduced-order model for high-dimensional microphysics, delivering >5,000× greater amortized integration throughput than the reference simulator.
- Built a permutation-invariant VAE that compressed up to 15,000 raw inputs into a 10-coordinate state while reconstructing four physical outputs with 1.04% median symmetric error.
- Derived a scale-covariant Neural ODE forecasting 4 physical outputs over 48 h with 1.21% median symmetric error.
- Devised a frozen-decoder consistency objective and applied latent-noise augmentation to curb overfitting; the latter cut the 24-h latent train–validation MSE gap by 39%.
- Designed multistep BPTT training that cut mean and P99 rollout errors by 17.6% and 28% vs. single-step training.
- Replaced quadratic partition scans with O(1) maps, cutting encoder-decoder pipeline startup from >35 min to 4.75 min.
- Re-engineered NetCDF/Zarr I/O with shared-memory staging, coalesced writes, and CUDA prefetch, raising loader throughput 33× vs. lazy loading while keeping the measured data-wait fraction below 7%.

➤ ***Particle-Resolved Simulation Engine (PartMC)*** | *Computational Modeling Engineer*

- Extended PartMC to simulate cloud ice formation, ice crystal growth, and nonlinear aerosol–droplet–ice interactions.
- Reduced a particle-interaction bottleneck from $O(n^3)$ to $O(n)$ through mathematical derivation and formal proof, making analyses of 10,000+ particles computationally tractable.
- Implemented Binned Tau-Leaping, accelerating simulations 8× with <0.1% numerical deviation.
- Architected and maintained PartMC's freezing-process module while contributing to the broader C++/Fortran simulation engine on HPC.
- Containerized reproducible cross-platform releases with Docker and built GitHub Actions CI/CD with Codecov, raising unit-test coverage above 94%.

ML Intern, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences, Beijing, China **2022**

➤ ***Air Quality Prediction*** | *Machine Learning Engineer*

- Built an end-to-end LSTM O_3 prediction system with a unified API that automated ingestion and normalization of meteorological, hydrological, and agricultural data.
- Engineered layer-wise transfer learning for sparse-data regions, cutting RMSE by ~51% vs. baseline models.
- Reconstructed seven years of continuous 0.1° O_3 fields across China from sparse ground-station observations.

Research Assistant, Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China **2020 – 2022**

➤ ***Bayesian Inversion System Development*** | *Computational Modeling Engineer* | *Project Lead*

- Led end-to-end development of BIS-v6.0, a modular 4D-Var system for estimating emissions from observations. Its decoupled OOP architecture enabled new algorithms with minimal refactoring.
- Implemented temporal-consistency constraints, improving R^2 by 21% over the legacy time-independent baseline.
- Developed a parallel sparse-matrix solver that ran ~460× faster than the legacy serial dense-matrix baseline.
- Built registry APIs and a Cron-orchestrated Linux ETL pipeline that standardized 19 emission inventories and automated simulation-to-dashboard workflows.

SELECTED PUBLICATIONS

- **Tang et al.** Learning Aerosol Coagulation Dynamics in Latent Space with a Scale-Covariant Neural ODE. *(arXiv)*.
- **Tang et al.** The impact of aerosol mixing state on immersion freezing: insights from classical nucleation theory and particle-resolved simulations. *Atmospheric Chemistry and Physics*, 26, 9221-9255 (2026).